

Spectrally Sharp Plasmon Resonances in the Near Infrared: Subwavelength Core-shell Nanoparticles

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Subwavelength plasmonic nanostructures have demonstrated many interesting phenomena and applications in recent decades. However, plasmonic resonance and associated properties, such as strong field enhancement, are limited to the visible regime due to the quasistatic nature of subwavelength plasmonic nanoparticles. Here, we numerically demonstrate spectral resonance tuning of subwavelength nanoparticles with a high-index core and a plasmonic shell while maintaining a sharp spectral peak. These core-shell-based nanoparticles have predominantly nonradiative loss and strong near-field enhancement due to their subwavelength feature size. This scheme can be further extended for magnetic and chiral meta-atoms. We suggest that these features of subwavelength core-shell-based nanoparticles are advantageous for many interesting applications including photothermal devices, enhanced light-matter interaction, and photonic circuitry.

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I. INTRODUCTION

Plasmonic nanoparticles have demonstrated many interesting phenomena or applications, including surface-enhanced Raman spectroscopy [1], the photothermal effect [2], refractive-index sensing [3], and enhanced light-matter interaction mediated at the nanoscale [4], owing to strong near-field enhancement [1,2] and/or strong and spectrally sharp resonant scattering [3].

An interesting feature of plasmonic nanoparticles is that their sizes can be much smaller than the resonance wavelength. On the contrary, the classical antenna scaling law states that antennas at resonance should be longer than half of the resonant wavelength [5], so antennas usually have a spectral resonance on a similar scale to the feature size. However, the localized-surface-plasmon-resonance (LSPR) wavelengths of subwavelength plasmonic nanoparticles are fixed around $\omega_{\text{LSPR}} \sim \omega_p/\sqrt{3}$ determined by the plasmonic material. Spectral resonance tuning has been demonstrated using size or shape tuning for single particles [6] and interparticle coupling [7]. Nevertheless, the spectral resonance of subwavelength nanoparticles cannot easily be engineered while maintaining a sharp spectral resonance peak and

usually involves spectral broadening and strong spectral deformation.

In this work, we numerically demonstrate spectral resonance tuning of subwavelength nanoparticles using the LSPR from the high-index dielectric core and the plasmonic shell. The physical origin of the LSPR mode is discussed using simple spherical core-shell systems and the plasmon-hybridization model. Throughout this work, we address the spectral resonance tuning through the inclusion of a high-index dielectric core into the plasmonic particle as the “core-shell (CS) scheme,” and the Si-core Ag-shell (Si@Ag) is specifically discussed (Fig. 1). The CS scheme also works for nonspherical particles, which we demonstrate using meta-atoms with magnetic dipole radiation and chirality. These CS-based nanoparticles have novel features: a sharp resonance peak in the near-infrared (NIR) regime, predominantly nonradiative loss, and strong near-field enhancement, which allow many interesting phenomena or applications including the photothermal effect in the NIR regime, subwavelength optical components for compact nanophotonic circuitry, and subwavelength nanoantenna with a resonance wavelength that is much shorter than the feature size.

II. MECHANISM OF CORE-SHELL SCHEME

We first analyze an already widely studied system of a dielectric core and a lossless plasmonic-shell sphere, for which the eigenfrequencies of LSPR in the quasistatic limit

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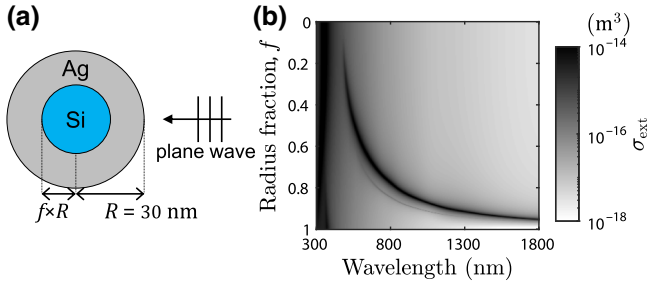


FIG. 1. (a) The schematics of a spherical high-index dielectric core and a plasmonic shell (Si@Ag) under plane-wave incidence. (b) The extinction cross section of the spherical core shell with a different radius fraction f and outer radius $R = 30$ nm.

are as follows [8]:

$$\omega_{n,\pm}^2 = \frac{\omega_p^2}{2} \frac{2(1 - \chi_n) + \Omega \pm \sqrt{\Omega^2 - 4\epsilon_c\epsilon_h(1 - \chi_n)^2}}{(1 - \chi_n)(1 + \epsilon_c\epsilon_h) + \Omega}, \quad (1)$$

where $\chi_n = f^{1+2n}$, $\Omega = \chi_n(\epsilon_c + \epsilon_h) + \epsilon_cn/(n+1) + \epsilon_h(n+1)/n$, f is the radius fraction between the shell radius and the core radius, n is the multipole order, which we consider to be $n = 1$ for the predominant dipole mode while ignoring higher-order multipole modes, and $\epsilon_{c,h}$ are the permittivities of the core materials and the host media, respectively. For Eq. (1), the plasmonic shell is modeled using the Drude model without damping: $\epsilon_s(\omega) = 1 - \omega_p^2/\omega^2$.

Specifically, when the core and host materials are vacuum [9], Eq. (1) is simplified to the well-known expression derived from plasmon-hybridization theory [10], which is the electromagnetic analog of molecular-orbital theory [Fig. 2(a)]. The hybridization theory describes the optical properties of composite plasmon structures having one or more element, with the hybridization of the various plasmon modes supported by each element. Therefore, the splitting of eigenfrequencies into $\omega_{\pm}$ is explained instructively by plasmon-hybridization theory in terms of bonding (symmetrically coupled plasmon mode, ω_-) and antibonding (antisymmetrically coupled plasmon mode, ω_+) states of sphere and cavity modes, which mainly depend on the plasmons of the outer and inner surfaces, respectively.

Note that the eigenfrequencies of this system at the quasistatic limit do not depend on the outer radius but on the radius fraction f [Fig. 2(c)]. This is also a feature of plasmon-hybridization theory [Fig. 2(b)]. We further discuss the effect of the outer radius and the effect of the core-shell materials in Secs. S1 and S2, respectively, of the Supplemental Material [11–13].

Using the CS scheme, predominantly nonradiative loss in the NIR regime can be achieved. It is known that nonradiative loss increases strongly as the size of the plasmonic

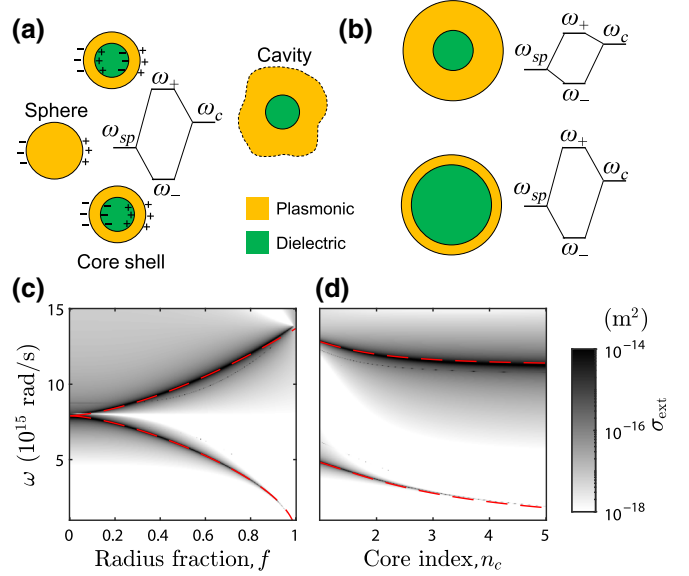


FIG. 2. (a) The schematics of the energy diagram for bonding (ω_-) and antibonding (ω_+) eigenenergy states due to hybridization of the sphere and cavity modes. (b) The energy diagram of the hybridized states for different shell thicknesses. (c), (d) Extinction cross sections of core-shell nanospheres calculated using the T-matrix method (black and white mapping) and the eigenfrequency calculated using Eq. (1) (red dashed lines) by varying (c) the radius fraction f and (d) the core index n_c . The Ag shell is modeled using the Drude model without damping, with $\omega_p = 1.3697 \times 10^{16}$ rad/s, and the dielectric core in (c) has a nondispersive refractive index of $n_c = 2$. The outer radius R is 10 nm and the radius fraction f is 0.8 for (d).

nanoparticles increases; therefore, it is difficult to achieve strong nonradiative loss in the NIR regime. To red shift the spectral resonance, the size of the plasmonic nanoparticles has to increase, but this rapidly increases the radiative damping of the nanoparticles [Fig. 3(a)]. On the other hand, the size of the CS nanoparticles can be kept constant and only the thickness of the metallic layer needs to decrease. Using the CS scheme, predominantly nonradiative damping in the NIR regime can be observed for small CS nanoparticles ($R = 30$ nm) [Fig. 3(b)].

Note that the demonstrated CS nanostructures have very thin plasmonic layers, of thickness possibly less than 5 nm. Such a system cannot be accurately described using the local dielectric function. Nonlocality has been treated by the hydrodynamic model [14] for a high-index dielectric core and a plasmonic shell [15] and it is known that the antibonding mode is strongly affected by nonlocality [16], whereas the bonding mode, which is of our interest in this work, is actually not strongly affected by nonlocality. We discuss the effect of nonlocality further in Sec. S3 of the Supplemental Material [11] and we proceed with a local description throughout this work using realistic optical parameters of Ag [17] and Si [18].

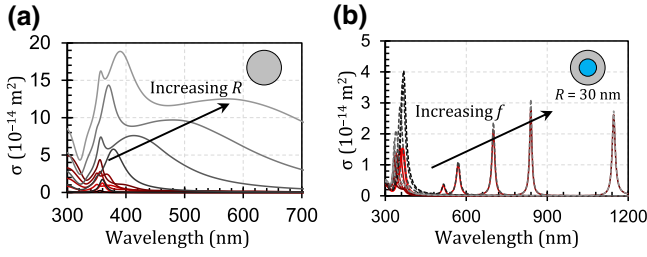


FIG. 3. Absorption (red) and extinction cross sections (black) of (a) a Ag sphere and (b) a Si@Ag core shell. (a) The Ag spheres have radii of $R = 10, 20, 40, 60, 80$, and 100 nm . (b) The CSs have radius fractions of $f = 0, 0.3, 0.5, 0.7, 0.8$, and 0.9 , and a radius of $R = 30 \text{ nm}$.

III. CORE-SHELL-BASED META-ATOMS

Metamaterials, or artificially structured materials, comprise another major application of plasmonic nanostructures. Plasmonic meta-atoms have demonstrated many exotic phenomena that are not easily achievable using bulk natural materials [19]. Such phenomena include structural coloring [20], the negative refractive index [21], and strong artificial chirality [22].

By introducing the CS scheme to these structured meta-atoms as an additional degree of freedom, the properties of the meta-atoms can be spectrally modified to exhibit a working wavelength that is much larger than their feature sizes. In this section, we demonstrate that this scheme works not only for simple extinction but also for magnetic dipole radiation and artificial chirality of meta-atoms.

We first compare LSPR-wavelength tuning using single nanorods and nanorods with the CS scheme. The resonance of a single Ag nanorod can be tuned to the IR regime by increasing the length of the nanorods [Fig. 4(a)]. However, as the nanorods become longer, radiative decay becomes dominant over nonradiative decay [23,24]. This increased radiation damping results in a broadened line width of LSPR as well as reduced local field enhancement [25]. On the contrary, Si@Ag CS nanorods demonstrate red-shifted extinction without spectral broadening; in fact, the spectra become even sharper [Fig. 4(b)]. The extinction of the CS nanorod does not increase as rapidly as the extinction of the Ag nanorod as the LSPR-wavelength red shifts, because the overall size of the Ag nanorod grows, whereas that of the CS nanorod remains fixed. It is also interesting to note that the scattering (red) is almost equal to the extinction (black) for the Ag nanorod, whereas intrinsic absorption can be observed for the CS nanorod. Although the absorption is not as dominant as for small particles [Fig. 3(b)], the CS system can exhibit noticeable nonradiative loss compared to a simple plasmonic system. We also quantitatively compare a Ag nanorod and a CS nanorod with the same resonant wavelength

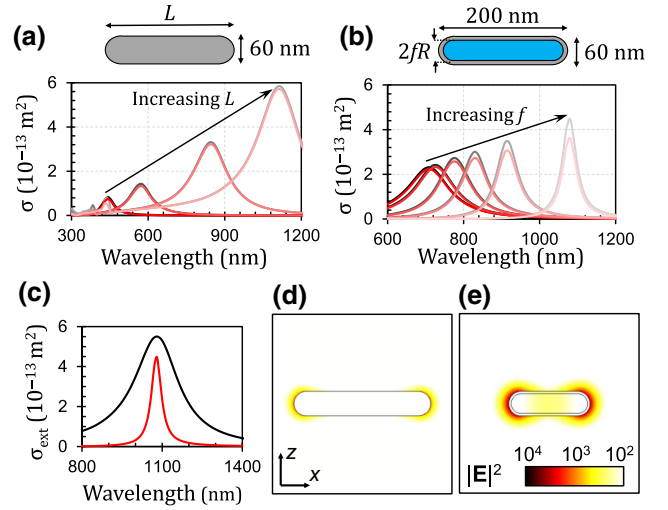


FIG. 4. Spectrally red-shifted extinction (black) and scattering cross sections (red) of (a) a Ag nanorod due to increasing L with $R = 30 \text{ nm}$ and $L = 100, 150, 250$, and 350 nm , and (b) Si@Ag CS nanorods due to increasing the radius factor f with $R = 30 \text{ nm}$, $L = 200 \text{ nm}$, and $f = 0, 0.3, 0.5, 0.6, 0.7$, and 0.8 . The incident plane wave is polarized parallel to the nanorod axis and is propagating normal to the nanorod axis. (c) The extinction spectra of a single Ag nanorod with $L = 337 \text{ nm}$ (black) and of Si@Ag CS nanorods with $L = 200 \text{ nm}$ and $f = 0.8$ (red). (d),(e) The electric field intensity near (d) a Ag nanorod and (e) a CS nanorod on a logarithmic scale at the resonant wavelength of 1080 nm .

of 1080 nm , where the Ag nanorod has $L = 337 \text{ nm}$ and the CS nanorod has $L = 200 \text{ nm}$ and $f = 0.8$. The CS nanorod has a sharper spectral resonance [Fig. 4(c)]; the calculated full-width-at-half-maximum values for each nanorod are 190 nm and 40 nm , respectively. In addition, the CS nanorod has stronger local field enhancement compared to Ag nanorods [Figs. 4(d) and 4(e)], due to reduced radiative damping while maintaining similar extinction.

Subwavelength nanoparticles generally exhibit a predominant electric dipole mode, but subwavelength-structured meta-atoms have demonstrated a magnetic dipole mode from a split-ring resonator [26], a plasmonic double bar [27], and high-index dielectric nanoparticles [28]. Such meta-atoms with a magnetic dipole mode can demonstrate negative-refractive-index media [29], directional scattering [30], and metasurfaces with high efficiency [31]. We demonstrate that this CS scheme of spectral tuning also works for magnetic dipole radiation using the subwavelength Si@Ag CS nanoparticles of a split-ring resonator (SRR) and vertical-double-bar (VDB) configurations. The magnetic meta-atoms discussed here already have red-shifted resonances due to plasmon coupling, but are even more red-shifted using the CS scheme. The SRR exhibits a magnetic dipole mode oriented in the z direction due to the current loop along the wire in the

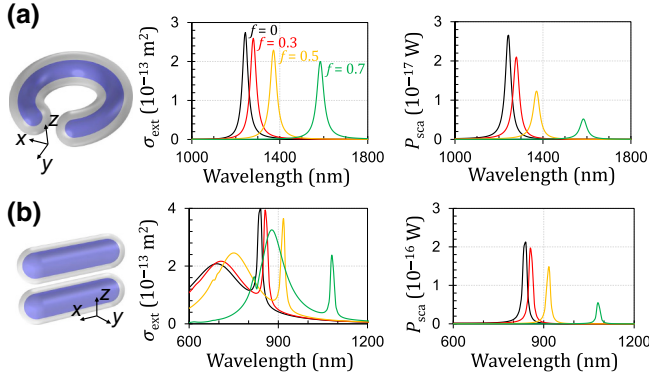


FIG. 5. Magnetic meta-atoms. The red-shifted extinction and magnetic dipole scattering power of (a) a split-ring resonator and (b) a double bar. Radius factors of $f = 0$ (black), 0.3 (red), 0.5 (orange), and 0.7 (green) are presented. The incident plane wave is polarized in the x direction and is propagating in the $+z$ direction. The geometric parameters are as follows: (a) major radius 50 nm, minor radius 20 nm, and split angle 60° ; (b) length 200 nm, outer radius 30 nm, and gap distance between outer shells 10 nm.

x - y plane. A Ag SRR ($f = 0$) already has its resonance in the NIR regime (1240 nm), but the resonance peak red shifts up to 1580 nm for $f = 0.7$ [Fig. 5(a)]. Unlike the CS nanorods [Fig. 4(b)], the extinction maxima decrease as f increases and the magnetic dipole radiation decreases even faster than the extinction as f increases. A VDB exhibits a magnetic dipole mode oriented in the y direction when excited by a plane wave polarized in the x direction and propagating in the z direction, which introduces phase delay in the z direction. A Ag VDB ($f = 0$) has a magnetic dipole mode at 840 nm and its magnetic resonance shifts to 1080 nm at $f = 0.7$ [Fig. 5(b)]. Interestingly, the electric dipole mode in the VDB (broad main peak) becomes stronger and sharper as f increases, whereas the magnetic dipole mode becomes weaker and broader. Compared to homogeneous Ag meta-atoms, CS meta-atoms exhibit more complicated modes, such as the Fano-like spectra around 820 nm for $f = 0.7$ of the VDB [Fig. 5(b)], but the general features are retained.

Strong artificial chirality far exceeding that of natural molecules has been demonstrated for the plasmonic nanoparticles of a helical assembly of nanospheres [32], twisted nanorods [33], a helix [34], and a continuous nanocrystal [35,36]. These chiral meta-atoms may demonstrate negative-index media [37] and efficiently fabricate color filtering by polarization rotation in chiral media [32, 36]. However, subwavelength chiral plasmonic nanoparticles usually have their chiral responses near LSPR wavelengths in the visible regime [38]. Here, we deploy the CS scheme to demonstrate spectral tuning of the chiral response using subwavelength chiral Si@Ag CS nanoparticles: a helical assembly, twisted nanorods, and a single

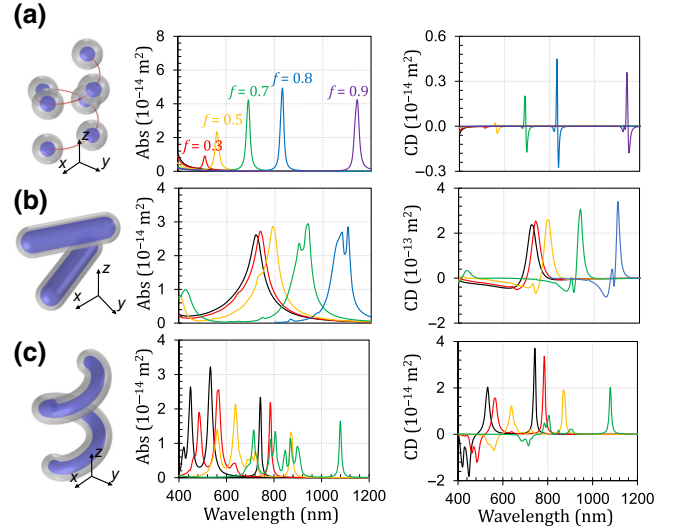


FIG. 6. Chiral meta-atoms. The red-shifted CD and Abs of (a) the helical assembly, (b) the twisted nanorods, and (c) the helix. Radius factors of $f = 0$ (black), 0.3 (red), 0.5 (orange), 0.7 (green), 0.8 (blue), or 0.9 (purple) are presented. The geometric parameters are as follows: (a) major radius 40 nm, minor radius 20 nm, pitch 70 nm, 1.5 turns, and seven particles; (b) radius 30 nm, length 200 nm, distance 50 nm, and twist angle 45° ; (c) major radius 20 nm, minor radius 10 nm, pitch 50 nm, and 1.5 turns.

helix. To characterize the chiral and achiral responses of these nanoparticles, we use circular dichroism (CD) and average extinction (Abs) defined as $\sigma_+ - \sigma_-$ and $(\sigma_+ + \sigma_-)/2$, respectively, where the $\sigma_\pm$ are the extinction cross sections due to the left- and right-circularly polarized light propagating in the $+z$ direction. The Ag helical assembly ($f = 0$) exhibits CD and Abs resonance near 400 nm, but this resonance shifts up to 1100 nm for $f = 0.9$ [Fig. 6(a)]. The extinction and CD of the bonding mode increases up to $f = 0.8$. Ag twisted nanorods ($f = 0$) have CD and Abs resonances at 700 nm, which red shift up to 1100 nm for $f = 0.8$ [Fig. 6(b)]. The Ag single helix ($f = 0$) has CD and Abs resonances at 750 nm, which red shift up to 1100 nm for $f = 0.7$ [Fig. 6(c)]. The bonding modes remain sharp and strong, but distorted modes are observed for the CS configuration of the single helix and the twisted nanorods.

IV. POTENTIAL APPLICATIONS

These CS nanoparticles have an absorptive loss that dominates the radiative loss compared to larger plasmonic nanoparticles (Fig. 3) and in the broadband NIR range [39]. This feature possibly allows the CS nanoparticles to have advantages in applications based on heat generation and in thermal devices. For instance, photoacoustic bioimaging based on the photothermal effect utilizes the absorption loss of plasmonic nanoparticles, which

rapidly generates heat and also generates ultrasonic sound [40]. The excitation light should be in the visible regime to match the LSPR resonance of the small plasmonic nanoparticles. The CS scheme moves the resonance wavelength to the NIR regime, which is the second biological window (1000–1400 nm), without increasing the particle size and simultaneously the radiative loss. The small size allowed by the CS scheme may itself be advantageous in intercellular bioimaging applications. This CS scheme may also be applied if some biomolecules are vulnerable to excitation light in the visible regime.

In addition, the CS scheme can potentially be used to realize subwavelength scattering systems with sharp and strong resonances in the NIR regime. Due to their small particle sizes, the CS scheme allows sharper resonances and stronger local-field enhancement. The strong local-field enhancement is especially beneficial for surface-enhanced infrared-absorption spectroscopy, which relies on local-field enhancement near plasmonic nanostructures to enhance molecular-vibrational absorption in the NIR regime.

Plasmonic nanoantennas can strongly modify nearby molecular dynamics [5,41], where the spontaneous emission rate can be strongly enhanced [42] and the emission direction can be controlled [43]. In addition, the optical modes of nanoantennas and molecular-vibrational modes can be coupled. The CS scheme may be utilized as a platform to study resonant coupling between subwavelength nanoantennas and molecular-vibrational modes in the NIR regime.

Photonic circuitry may possibly overcome the switching rate of electronic logic circuits, which is limited to around the gigahertz region. However, the elements of the photonic circuits are large, so it is difficult to realize dense photonic circuits. Plasmonics could make the photonic components much smaller by supporting subwavelength modes [44,45], but plasmonic nanostructures usually have their responses around the LSPR wavelength, which is much shorter than the telecom wavelength of around 1500 nm. We believe that these CS-based nanostructures could substantially reduce feature sizes, possibly realizing compact and dense photonic components. Also, because the structural feature sizes are much smaller than the working wavelengths, efficient designs of photonic components based on an effective metamaterial approach could be used. Considering that CS-based nanostructures behave in a similar way compared with the fully plasmonic cases ($f = 0$), except that their responses become red shifted, we can design CS-based nanostructures using the design rules of plasmonic structures obtained from decades of experiences, as we demonstrate using the meta-atoms. Previously demonstrated structures, such as an Al layer on Si blocks with sharpened and red-shifted resonances [46] and hybrid plasmonic waveguides [47], could be related to the CS scheme.

V. CONCLUSION

In conclusion, we discuss the CS scheme as a way of spectrally red shifting the sharp LSPR peak of subwavelength nanostructures. We briefly discuss the mechanism of the spectral red shift coming from the bonding states of the particle and cavity modes due to plasmon-hybridization theory. The CS scheme allows a sharp spectral resonance in the NIR regime with strong near-field enhancement due to reduced radiative loss even using smaller particle sizes. This CS scheme is also confirmed to work for meta-atoms with magnetic dipole radiation and chirality. Based on the above observations, we suggest several potential applications of these CS systems, including bioimaging in the NIR regime based on the photothermal effect, enhanced light-matter interaction in the NIR regime due to strong local-field enhancement, and ultracompact photonic circuitry.

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